Technical review

Item #3 · original.pdf · Run ID: 20261113T232756Z-94fc69

Path	R:\evidence\pdf\original.pdf
Size (bytes)	1648
Last modified (UTC)	2026-11-12T06:00:00Z
Extension	.pdf
Detected type (signature)	PDF document
Status	No issues found
Case	RV-SAMPLE-001
Evidence ID	E0020
Case registry	Verified: identical to the registered evidence
Registered at intake (UTC)	2026-11-13T23:27:52.795Z

Hashes

sha256	`7446db6b551798dc8389da3e84e90bc5caf40c3008b71fc19e56afd13164c743`
md5	`8d2b5163663d6eaf0ff751d364efe5d6`

Findings

None

Revision history

#	Byte range	Saved at (Info)	Saved at (XMP)	Producer	Time since previous	Objects added / changed / deleted	Pages changed	Signed by (field)
1	0–1648 (1648 B)	2026-02-16T15:30:00Z	—	ReportLab PDF Library - (opensource)	—	8 / 0 / 0	—	—

Timeline

#	Saved at (UTC)	Time since previous	Producer
1	2026-02-16 15:30:00	—	ReportLab PDF Library - (opensource)

Dates are those stored in the file (Info dictionary, else XMP); the software that saved it can set them freely. Without usable dates the revisions are spaced evenly.

Signatures

None

The signer certificate, its chain and its revocation status were not validated: a mathematically valid signature does not show who controlled the key.

Fonts

Font	Subset prefix	Embedded	First revision	Object
Helvetica	—	no	1	2 0
Helvetica-Bold	—	no	1	3 0

Producer fingerprints

Role	Database	Version	Entries	Status	SHA-256
starter	rakvor-starter	2026.10.21	37	loaded	`0b8df17b3ba7f33932cfc072932f198cdfdba98ad1332b33ac84c5c2f9068795`
local	—	—	—	absent	—

#	Tool	Category	Matched on	Value found	Traits not matching
1	ReportLab PDF Library	pdf_library	producer	ReportLab PDF Library - (opensource)	—

Visual comparison

Status	Not applicable
Reason	single revision
Pages changed	0

No difference image was written.

Parser warnings

None

The complete list of parser warnings is in warnings.json in this item's folder (its digest is in the audit log and in the extracted files table).

Extracted files

warnings.json (parser_warnings, 123 B)	md5: 251720985f0ba1c888116538bc3da921 sha256: 9dfdca46175986eda2dbcd6912894e5d123f4f0d267ee0e0b018d8d96fc149ae
objects.json (objects_dump, 2824 B)	md5: 532955e08f79ac83a25130173cfdbe6f sha256: b10efd15e93ee257ca35ac76fce3d26fd65479aad64b5113e224fb57baf9f7a2

Method

Each file was opened read-only and hashed in a single pass with the selected algorithms. PDF files were then parsed without being modified: every saved revision, its dates and producer, digital signatures, fonts and producer fingerprints were recorded, and pages of consecutive revisions were compared visually. Every action was written to a hash-chained audit log.

Audit trail

Audit entries	#29 – #35
Log folder	`R:\case\runs\rv-pdfauth\20261113T232756Z-94fc69\logs`
Digital signature	This PDF is not digitally signed.

Limitations

- Type detection uses a limited table of common file signatures; unknown types are reported as unknown, not as suspicious.
- Modification times come from the file system of the examined copy and may differ from the original source.
- A result without findings does not prove that a file is authentic or unaltered before acquisition.
- A trusted timestamp proves the audit log existed in this exact form at the stated time; it does not prove that the examined copy matches the original source.
- The absence of findings does not prove that a PDF is authentic or unaltered: a document can be forged before it is saved, or re-created so that no trace remains.
- When a PDF is printed to PDF, flattened, optimised or regenerated, its earlier revisions and most editing traces are lost; only its last state can be examined.
- Producer fingerprints come from a limited database (vendor documentation and files generated by Rakvor); producer strings can be typed by hand, and unknown or spoofed values cannot be recognised.
- Digital signatures are checked for byte coverage and cryptographic integrity only; certificate chains, revocation and the signer's identity are not validated.
- The visual comparison between revisions renders only a limited number of pages per revision; pages beyond that limit are compared structurally only.